PROJECT REPORT

TASK 23: DUPLICATE DATA REMOVAL USING PANDAS

1. Abstract

This project demonstrates the identification and removal of duplicate records from a dataset using Python and Pandas. A sample student performance dataset containing repeated records is used to identify duplicates, calculate their count, remove repeated entries, and compare the number of records before and after cleaning.

The project introduces an important data-cleaning technique that improves dataset quality before performing data analysis or machine-learning modelling.

2. Introduction

Duplicate data occurs when identical records are stored more than once in a dataset. These repeated observations can affect the accuracy of analysis, statistics, and machine-learning results.

For example, if the same student's marks are recorded multiple times, the analysis may incorrectly give that student additional importance.

Pandas provides functions such as `duplicated()` and `drop_duplicates()` to identify and remove duplicate rows efficiently.

This project uses a student performance dataset to demonstrate the duplicate-removal process in Google Colab.

3. Objectives

The main objectives are:

- To identify duplicate records in a dataset.
- To calculate the total number of duplicate entries.
- To display duplicate-related records.
- To remove duplicate records using Pandas.
- To verify that duplicates have been removed.
- To compare dataset sizes before and after cleaning.
- To understand the importance of duplicate removal in data analysis.

4. Tools and Technologies

- **Python:** Used to develop and execute the program.
- **Pandas:** Used to identify, count, and remove duplicate records.
- **NumPy:** Used for supporting data representation.
- **Google Colab:** Used as the Jupyter Notebook development environment.
- **CSV:** Used to store and export the original and cleaned datasets.

5. Dataset Description

The project uses a manually created student performance dataset containing 11 records and six columns.

The columns are:

Column	Description
student_id	Unique student identifier
name	Student name
department	Academic department
age	Student age
marks	Marks obtained
attendance	Attendance percentage

Three records have been intentionally repeated to demonstrate duplicate detection and removal.

The dataset is fictional and used only for educational purposes.

6. Methodology

The project follows these steps:

1. Import Pandas and NumPy.
2. Create a student performance dataset.
3. Insert intentional duplicate records.
4. Convert the dataset into a Pandas DataFrame.
5. Export the original dataset as a CSV file.
6. Load the CSV file using Pandas.
7. Identify duplicate entries using `duplicated()`.
8. Count the repeated records.
9. Display all records belonging to duplicate groups.
10. Remove duplicates using `drop_duplicates()`.
11. Verify the cleaned dataset.
12. Compare the record count before and after cleaning.
13. Export the cleaned dataset and summary.

7. Duplicate Identification

The Pandas `duplicated()` function identifies repeated rows by returning Boolean values.

By default, the first occurrence of a record is considered original, and subsequent identical occurrences are marked as duplicates.

The expression `df.duplicated().sum()` calculates the number of duplicate entries.

In this project, three duplicate entries are identified.

The expression `df.duplicated(keep=False)` displays all occurrences belonging to duplicate groups, including the original records.

8. Duplicate Removal

The `drop_duplicates()` function is used to remove repeated rows.

The project uses:

```
cleaned_df = df.drop_duplicates()
```

By default, the function retains the first occurrence of each identical record and removes the subsequent repeated entries.

The original DataFrame is preserved, while the cleaned records are stored in a separate DataFrame.

9. Results and Observations

The following results are obtained:

Description	Number of Records
Original dataset records	11
Duplicate entries identified	3
Records after cleaning	8
Remaining duplicate entries	0

The cleaning operation removes three repeated records and retains eight unique records.

All remaining records are verified using `duplicated().sum()`.

10. Before-and-After Dataset Comparison

Before cleaning, the dataset contains 11 records, including three repeated entries.

After removing duplicates, the cleaned dataset contains eight records.

The number of records removed is:

Records Removed = Original Records − Cleaned Records

Records Removed = 11 − 8 = 3

The results confirm that the duplicate-removal operation was successful.

11. Importance of Duplicate Removal

Removing unintended duplicate records is important because repeated observations can affect statistical calculations and analytical conclusions.

In machine learning, duplicate records can cause certain observations to receive excessive weight and may create problems during data preparation and model evaluation.

Duplicate removal helps maintain data consistency and improve the reliability of subsequent analysis.

12. Applications

Duplicate removal is useful in:

- Student information management systems.
- Customer databases.
- Employee records.
- Sales transaction datasets.
- Banking and financial records.
- Customer relationship management systems.
- Data preparation for machine learning.

13. Limitations

This project uses a small manually created dataset with exact duplicate rows.

In real-world datasets, records may be similar without being exactly identical. Identifying such records may require additional matching rules, such as comparing selected columns or using fuzzy matching techniques.

14. Future Enhancements

The project could be extended by:

- Identifying duplicates using selected columns.
- Detecting similar but non-identical records.
- Cleaning larger real-world datasets.
- Generating data-quality reports.
- Creating visualizations comparing original and cleaned datasets.
- Automating duplicate detection in regular data-processing workflows.

15. Conclusion

This project demonstrates how Python and Pandas can identify and remove duplicate records from structured datasets.

Using `duplicated()` and `drop_duplicates()`, three duplicate entries were identified and removed from an original dataset containing 11 records.

The cleaned dataset contains eight unique records, demonstrating a successful basic data-cleaning operation.

This exercise provides a practical foundation for maintaining data quality before performing data analysis or machine-learning modelling.

16. References

1. Pandas Documentation: https://pandas.pydata.org/docs/
2. Pandas `duplicated()` Reference: https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.duplicated.html
3. Pandas `drop_duplicates()` Reference: https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.drop_duplicates.html

18. Project Links

GitHub Repository: https://github.com/venky1318/Duplicate_Data_Removal